

The multiplicative group modulo p : a primer for dummies

What is a group?

→ $(G, \circ)$

⇒ G = set of elements (group members)

⇒ $\circ$ = operation (group operation)

→ **4 properties:**

⇒ **Closure:** for any g_1, g_2 :

$g_x = g_1 \circ g_2$ must be a group member

⇒ **Identity:** there is a group member I such that
 $g \circ I = I \circ g = g$

⇒ **Inverse:** for any g , there is g^{-1} such that
 $g \circ g^{-1} = I$

⇒ **Associativity:** for any g_1, g_2, g_3 :
 $(g_1 \circ g_2) \circ g_3 = g_1 \circ (g_2 \circ g_3)$

→ **If also commutative → Abelian Group**

The $\mathbb{Z}_p^*$ group

→ multiplicative group modulo prime p

⇒ Set of $p-1$ elements $\{1, 2, \dots, p-1\} \rightarrow$ finite group

⇒ Multiplicative = we only care about multiplications mod p !!

→ Forget the sum, here (otherwise you would have a field $\mathbb{F}_p$)

→ Group properties:

⇒ Closure: obvious

⇒ Identity: obvious

⇒ Associativity: obvious

⇒ Commutativity → Abelian group → also obvious

⇒ What about inverse???

→ If mod N , then x has inverse if and only if $\gcd(x, N) = 1$

→ If $N = p = \text{prime}$, then all elements have inverse!

» Note that 0 is not an element of the group!

Example: $\mathbb{Z}_{11}^*$

→ **Elements:** $\{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$

→ **Inverses:**

⇒ $1 \rightarrow 1$

⇒ $2 \rightarrow 6$ $6 \rightarrow 2$

⇒ $3 \rightarrow 4$ $4 \rightarrow 3$

⇒ $5 \rightarrow 9$ $9 \rightarrow 5$

⇒ $7 \rightarrow 8$ $8 \rightarrow 7$

⇒ $10 \rightarrow 10$ (analogous to -1)

→ **How to compute inverses for large groups?**

⇒ Extended euclidean algorithm

Back to multiplicative groups: exponentiation

→ $\mathbf{x}^k = \mathbf{x} \circ \mathbf{x} \circ \mathbf{x} \circ \dots \circ \mathbf{x}$ (**k times**)

→ **Generator of group of order m**

⇒ exists g such that

$\{g^0, g^1, \dots, g^{m-1}\} =$ all m group members

→ **Prime-order group:**

⇒ If m is prime, any member is generator

→ Except the identity

→ **Is $\mathbf{Z}_p^*$ a prime order group? NO!!**

⇒ $|\mathbf{Z}_p^*| = p-1$ CANNOT be prime

→ p is prime → p-1 is even

Example: $\mathbb{Z}_{11}^*$

→ **Elements:** {1,2,3,4,5,6,7,8,9,10}

→ **Generators?** $\{g^1, g^2, g^3, \dots, g^{10}\} = ?$

⇒ $g=2 \rightarrow \{2,4,8,5,10,9,7,3,6,1\}$

OK, generator

⇒ $g=3 \rightarrow \{3,9,5,4,1,3,9,5,4,1\}$
Subgroup order 5

NO!

⇒ $g=4 \rightarrow \{4,5,9,3,1,4,5,9,3,1\}$
Subgroup order 5

NO!

⇒ $g=5 \rightarrow \{5,3,4,9,1,5,3,4,9,1\}$
Subgroup order 5

NO!

⇒ $g=6 \rightarrow \{6,3,7,9,10,5,8,4,2,1\}$

OK, generator

⇒ $g=7 \rightarrow \{7,5,2,3,10,4,6,9,8,1\}$

OK, generator

⇒ $g=8 \rightarrow \{8,9,6,4,10,3,2,5,7,1\}$

OK, generator

⇒ $g=9 \rightarrow \{9,4,3,5,1,9,4,3,5,1\}$
Subgroup order 5

NO!

⇒ $g=10 \rightarrow \{10,1,10,1,10,1,10,1,10,1\}$
Subgroup order 2

NO!

→ **Take home:**

⇒ either g is a generator

⇒ Or generates a SUBGROUP → order = factor of $|G|$

⇒ **And $\mathbb{Z}_p^*$ as well as all subgroups are cyclic!**

Strong primes

→ Prime p such that
 $p = 2q + 1$
being q also prime!

→ Order of $\mathbb{Z}_p^*$: $p-1$
 $p-1 = 2q$

→ Hence, any member x (except 1 and $p-1$)
either

1. Generates the whole group, or
2. Generates subgroup of prime order q

→ **Both large if p and q large!**

⇒ Note the difference when $\mathbb{Z}_p^*$ uses «just» a large prime p :
 $p-1$ can factor down in small numbers!

Quadratic residue subgroup

➔ $x \in Z_p^*$ is a **quadratic residue** if it admits **square root in Z_p^***

⇒ i.e., there exists a such that $a^2 \bmod p = x$

➔ **QR form subgroup of order $\frac{p-1}{2}$**

⇒ $2 \rightarrow 1$ mapping: $\begin{matrix} x \\ p-x \end{matrix} \rightarrow x^2$ Indeed, $(p-x)^2 \bmod p = p^2 - 2px + x^2 \bmod p = x^2$

➔ **QR test: legendre symbol**

⇒ $a \in QR$ if $a^{\frac{p-1}{2}} \bmod p = 1$ (otherwise -1)

➔ **Example for Z_{11}^* : $QR_{11} = \{1, 3, 4, 5, 9\}$**

⇒ If p strong prime, QR_p has prime order q !